🌿 ArecaLens — Comprehensive Project Summary

Explainable AI (XAI) & Knowledge-Grounded Agricultural Assistant for Arecanut Disease Detection

Full Engineering, Architecture & Verification Report • Antigravity AI

1. Executive Summary & Purpose

ArecaLens is an end-to-end Explainable Artificial Intelligence (XAI) screening aid and knowledge-grounded agricultural assistant designed for arecanut (*Areca catechu*) growers in Karnataka and Kerala, India. The application targets critical fungal diseases—predominantly **Mahali / Koleroga** (fruit rot caused by *Phytophthora meadii*) and **Yellow Leaf Disease (YLD)**.

Unlike generic crop apps that risk farmer livelihood through hallucinated pesticide doses or false reassurances, ArecaLens enforces strict protective principles: **high-confidence thresholds** (categorizing borderline scans as "Unsure"), **zero chemical dosage invention**, and **complete offline determinism**.

> **Engineering Core Tenet**
>
> *"ArecaLens includes a knowledge-grounded retrieval assistant (not a generative one) that answers from cited, expert-review-pending sources with strict offline safety."*

2. System Architecture & Technology Stack

Component	Technology	Key Responsibilities
Web Frontend	React 19, Vite, Vanilla CSS	WCAG AA accessible interface, interactive CAM heatmap slider, 4-language toggle (EN, KN, HI, ML), local 8-scan history.
Backend API	FastAPI, Uvicorn, Python 3.13	Asynchronous REST endpoints, threadpool inference, zero message logging, unified static production serving on port 8000.
Computer Vision & XAI	TensorFlow 2.21, Keras, TFLite	MobileNetV3Small classification, 224x224 half-pixel bilinear preprocessing, deterministic Class Activation Mapping.
Offline Chatbot	Python Standard Library	Deterministic TF-IDF, synonym mapping, 27 vetted FAQ entries & disease guides. Completely offline with zero hallucinations.
Online LLM Proxy	OpenAI-compatible client	Optional server-configured RAG with Google Gemini / Groq; strict server-side regex post-checks blocking unvetted chemical doses.
Companion Mobile	Flutter (Dart)	Cross-platform 5-tab mobile application with camera capture, offline status indicator, and golden parity chat engine.

3. Computer Vision & Explainable AI (XAI) Pipeline

- **Target Classes & Protective Cutoffs:**
  - `Healthy_Leaf` (Confidence ≥ 0.95)
  - `Healthy_Nut` (Confidence ≥ 0.95)
  - `Mahali_Koleroga` (Fruit rot, Confidence ≥ 0.70)
  - `Yellow_Leaf` (Yellow Leaf Disease, Confidence ≥ 0.70)
  - *Scans scoring below thresholds are categorized as* ***"Unsure"*** *to prevent dangerous false reassurances.*
- **Deterministic Class Activation Maps (CAM):** Generates feature-weighted activation heatmaps (**CAM = ReLU(features × W_{eff})**) overlaid directly on raw photos to reveal visual lesion boundaries.
- **Preprocessing & Colab Parity:** Bilinear resize to 224x224 with exact TensorFlow half-pixel centers (float32 [0, 255]), mathematically validated against Google Colab baseline exports.
- **Local Privacy:** Retains up to 8 recent scans in browser `localStorage`; raw image uploads are never stored on server disks.

4. Multilingual Knowledge-Grounded Chatbot ("Ask" Tab)

- **100% Deterministic Grounding:** Standard-library engine (`backend/chat_engine.py`) answers strictly from verified sources (`chatbot_faq.json`, `disease_info.json`).
- **Result-Aware Context:** Features an "About your last result" chip linking screening outcomes to follow-up care without re-transmitting photos.
- **Mandatory Safety Guardrails:**
  - `_DOSE`: Refuses unverified spray quantities and concentrations; directs to agricultural officers and KVKs.
  - `_HEALTH`: Intercepts questions regarding human poisoning or ingestion, urging immediate emergency medical help.
  - `_CONFIRM`: Enforces that screening results are "suspected", never "confirmed".
  - `_OUT`: Politely declines unrelated queries (market prices, weather, other crops).
- **Multilingual Localization:** Localized answers and dynamic suggestion chips across English (`en`), Kannada (`kn`), Hindi (`hi`), and Malayalam (`ml`) with clear draft-translation notices.

5. Online "Ask More" Cloud Proxy & RAG Safety Post-Checks

- **Server-Grounded Context:** Automatically retrieves top-3 FAQ items and active class guidance as strict context for Google Gemini or Groq LLMs.
- **Strict Post-Generation Audit:** Scans model output; immediately discards and reverts to verified guide text if the model outputs unverified units (`ml`, `g`, `kg`, `litre`, `ppm`, `%`) or certainty terms (`confirmed`, `100%`).
- **Transparent Provenance:** Clearly badges responses as "*AI-generated • Verify with officer*" vs "*From the verified guide*".

6. Comprehensive Verification Matrix

Suite	Scope	Result	Status
Backend Pytest	FastAPI, CAM, Preprocessing, Chat API	83 passed, 0 failed (in 2.48s)	100% Pass
Frontend Vitest	AskTab, Sliders, History, Localization	30 passed across 8 test suites	100% Pass
Chatbot Dev Set	64 representative farmer questions	64 / 64 exact matches	100% Match
Chatbot Unseen Set	20 unseen, paraphrased questions	15/20 exact (75%), 20/20 safe outcomes	100% Safe
Chemical Doses Audit	All evaluation and test sets	Strictly 0 unverified doses generated	0.0% Hallucination
Credential Audit	Git history and working directory	0 keys or secrets committed	Clean

7. Companion Flutter Mobile Application

Located in the `mobile/` directory, a Flutter cross-platform application complements the web platform. It features local farmer onboarding, 5-tab navigation, camera/gallery scanning, offline network indicators, and a pure Dart chat engine with 100% golden parity to the backend logic.

8. How to Launch the Web Application

The single-port production server delivers both the frontend React application and the FastAPI backend:

```
# Quick Launch (Recommended):
Double-click start.cmd  (or run .\start.ps1 in PowerShell)

# Direct PowerShell Command:
.\backend\.venv\Scripts\python.exe -m uvicorn backend.main:app --host 127.0.0.1 --port 8000

-> Access the web interface at: http://127.0.0.1:8000
-> Access interactive API documentation at: http://127.0.0.1:8000/docs
```